space-suits have many individual components that use cutting edge technology. For example, there's the problem of keeping an astronaut from freezing or frying when, say, on the Moon. Since temperatures can reach extremes of between 120 degrees to minus 150 degrees Celsius, they need all the protection they can get. Though we may think that the technology to keep the temperature inside the suit bearable is similar to that in a fireman's suit, nothing could be further from the truth.

Cooling

Space suits used today have a special liquid cooling system, that maintain body temperatures by using a battery-operated pump to pass cold water through the tubes that line the suit fabric. This system was devised by Bill Elkins, an ex-NASA researcher at their Ames Research Center. He later founded Game Ready (**www.gameready.com**), a company that makes equipment to help professional athletes treat injuries.

Game Ready's cooling system is helping atheltes recover faster from muscle injuries

Their Game Ready Accelerated Recovery System helps in the treatment of sprains, bruises and post-surgery recovery. It is a computer-controlled unit that controls the flow of cold water to the specially designed flexible fabric wrap. It uses the same technology that's used in space-suits to control temperatures. The system is medically effective because of its ability to provide constant cooling and controlled compression. Soft tissue injuries now heal up to twice as fast because of this technology, and it's surely a boon to most professional or semi-professional athletes across the globe.

Filtration

Another very real problem when exploring space is the need for water. Since weight and volume have to be controlled on current spacecraft, storage of large amounts of water is impossible.

One method of conserving water is to make sure none is wasted. Distasteful as it sounds, the only way to conserve water in space is to recycle everything possible, including urine!

Argonide's filter system can remove bacteria and even poison from contaminated water.

Now, normal water cleansing methods, such as distilling, require large amounts of energy to boil the water, and this is again a big no-no on a space flight. Like water, energy needs to be conserved as well! The only remaining option is filtration, which uses little or no energy. However, developing a filter capable of purifying water takes some doing! Well, they did it!

Argonide (**www.argonide.com**), a company that researches advanced filtration, for use in space and back on Earth as well, developed nano-aluminium fibres that are only two nanometres thick. What's better, these fibres, trademarked as NanoCeram, displayed amazing bio-adhesive properties, which would help in removing bacteria and viruses. When woven into a filter, these fibres were electro-positively charged, thus making all electro-negative particles cling to the filter-bacteria, viruses, dyes and other impurities.

If that's not enough, these fibres also displayed properties that can stimulate new bone growth. Tests showed that when a NanoCeram net was grafted into a shattered bone, it attracted and retained bone cells, and caused accelerated, flawless growth of new bones!

Currently, Argonide is developing personal filtration devices that you will be able to use when out camping, or for military personnel to protect against contaminated or poisoned water.

Engines

NASA has long been researching better materials for construction of vital elements such as those in engines, or other moving parts. Several years ago, they planned to develop an aluminium alloy that had higher strength and resistance at high temperatures, such as those encountered in space. The project was conducted by Jonathan Lee, a structural engineer in the Materials, Processes and Manufacturing Department at the Marshall Space Flight Center, and PoShou Chen, a scientist from Morgan Research Corporation. When they succeeded, all the information was made public via the Internet, and public licensing made available.

Evinrude's Engine runs very silently and requires no maintenance for atleast 3 years

This information intrigued the people at Bombadier Recreational Products Inc., a manufacturer of outboard motors, who promptly contacted NASA in April 2002. The result is their brand new outboard motor called Evinrude E-TEC (**www.evinrude.com**), which is marketed as "every boater's dream!"

Actually, the engine is remarkable: it doesn't require any oil changes, is not affected by extreme temperatures, and doesn't need tune-ups for at least three years of normal use. What's more, it's also "whisper quiet", so you won't scare away the fish as you approach!

All this is thanks to NASA's aluminium alloy, which is 2.5 times stronger than regular cast aluminium. However, what's absolutely incredible is that this special alloy is produced at a material cost of less than Rs 90 per kilo!